

PHUMINAN KURODA

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SUMMARY

Fourth-year Information Technology student with hands-on experience building and deploying AI/ML systems end-to-end, including retrieval-augmented generation (RAG) pipelines, computer vision models, and a transformer language model implemented from first principles. Comfortable across the full stack, from data pipelines and model training to interactive web deployment. Strong academic record (GPAX 4.00) with prior experience supporting instruction as a teaching assistant.

EDUCATION

Bachelor of Science, Information Technology

2023 – 2027 (Expected)

Thai-Nichi Institute of Technology

- GPAX: 4.00 / 4.00

SKILLS

AI, ML & LLMs: PyTorch, Ultralytics YOLO, RAG Pipelines, ChromaDB, Ollama, Prompt Engineering

Languages & Databases: Python, C++, C#, Java, JavaScript, Oracle SQL, MongoDB

Cloud & Model Deployment: FastAPI, Docker, GitHub Actions, Google Cloud Platform, Render, Streamlit, Git

Analytics & Automation: Power BI, Looker Studio, Power Apps, Power Automate

Languages: Thai (Native), English (TOEIC 775), Japanese (Beginner)

PROJECTS

Hybrid RAG Document Q&A Chatbot

- Built a hybrid retrieval pipeline combining BM25 sparse search with dense embedding retrieval (ChromaDB), fused via Reciprocal Rank Fusion, **improving retrieval accuracy from 77.8% to 83.3% (+5.5 pts)** versus dense-only retrieval on an 18-question test set, while also **reducing average query latency from 23ms to 17ms**.
- Implemented dynamic PDF ingestion allowing users to upload and query custom documents in real time, with source-level citation on every answer.
- Deployed a fully local RAG pipeline (Gemma 3 via Ollama) with a provider-agnostic LLM layer, enabling switching between local and hosted inference without code changes.

PPE Compliance Detector & Production API

- Fine-tuned a YOLO26 object detection model on a construction-site PPE dataset to detect helmet compliance, **achieving 89.6% mAP@0.5 (53.4% mAP@0.5:0.95, 86.1% precision, 85.3% recall)** on the validation set.
- Wrapped the model as a production REST API (FastAPI) with input validation and automated tests; containerized with Docker and deployed via a GitHub Actions CI/CD pipeline to Render, **achieving 139.32ms average latency in production**.
- Diagnosed and fixed a channel-order (RGB/BGR) bug causing inconsistent detection results between the CLI and web app, restoring expected model accuracy.

GPT-Style Transformer Language Model from Scratch

- Implemented a decoder-only transformer (**~15.6M parameters**) from first principles in PyTorch, including scaled dot-product self-attention, multi-head attention, causal masking, and positional embeddings, without relying on high-level transformer libraries.
- Trained the model on a character-level fiction text corpus; identified overfitting from the validation loss curve (**best validation loss 1.41 around step 4000**) and implemented best-checkpoint saving to address it.
- Built an interactive text-generation demo to showcase the model's learned language patterns.

Classical ML: From-Scratch vs. Library Implementation Study

- Implemented Linear and Logistic Regression from first principles in NumPy (gradient descent, cross-entropy loss), matching scikit-learn's accuracy/MSE on both toy and real-world (Titanic) datasets.
- Benchmarked Decision Tree, Random Forest, and Gradient Boosting on the Titanic dataset, quantifying the bias-variance trade-off via train-test accuracy gaps (**0.041 to 0.113 across model types**).

EXPERIENCE

Teaching Assistant

Thai-Nichi Institute of Technology

- Mentored 30+ students in Java OOP, C#, and Database Systems labs, debugged code and explained OOP principles & SQL queries.
- Designed quizzes and midterm/final exam programming problems, evaluated student submissions alongside instructors.

CERTIFICATIONS & AWARDS

- Certificate of Academic Excellence — awarded 3 consecutive years (2024, 2025, 2026)
- TCS Academic Interface Program (AIP) — Artificial Intelligence & Cyber Security
- The Web Developer Bootcamp